

SENSORS

Force sensor

In early 2019, Alps Alpine developed HSFPAR004A force sensor, which fits into the tip of a stylus pen for tablet PCs



to enable accurate production of writing/drawing thickness corresponding to the pressure

applied. By achieving high sensitivity, high resolution, and high linearity, as well as great impact resistance, the HSFPAR004A was well received by many users.

Now, Alps Alpine has launched the new HSFPAR007A force sensor, a product that retains the basic performance capabilities of the HSFPAR004A. The new product is small enough to fit in the slim stylus pens used by smartphones, thus enhancing design flexibility and improving the user-friendliness of smartphone drawing applications.

The HSFPAR007A can detect stress as light as a zero stroke (0.01N), enabling handling precision in applications using robotic arms and other industrial equipment.

Alps Alpine

www.alpsalpine.com

Image sensor

Canon, a provider of innovative technologies for camera and vision capturing systems, has developed a single-photon avalanche diode (SPAD) image sensor with signal-amplifying pixels that are capable of capturing 1MP images.

SPAD image sensors are ideal for 2D cameras that capture and develop still images and videos in an extremely short time period. These sensors can also be



used in 3D cameras to get information about the distance between them and a subject as image data.

The sensor uses a method known as photon counting to realise a digital image resolution of 1MP. Additionally, the sensor has a shutter that allows simultaneous control of exposure for every pixel, making exposure time as short as 3.8 nanoseconds.

The sensor is capable of capturing up to 24,000 frames per second (FPS) with 1-bit output, thus enabling slow-motion capture of fast movement within an extremely short time frame.

Canon

in.canon

MEMS sensor

In order to avoid failure of the vehicle navigation system while venturing into unfamiliar territory, say an unknown city or road, Bosch



has introduced a MEMS sensor, SMI230. This highly precise sensor constantly

registers changes in the vehicle's direction and speed, evaluates the information, and transmits it to the navigation system. There, the information is combined with positional data from the global navigation satellite system (GNSS) and used for navigation.

Sudden interruptions in the GPS signal due to an overhead tunnel while driving can cause disarray in the navigation system. However, the SMI230 ensures that a vehicle's navigation system is free from such issues. Thanks to the combination of GPS position and sensor information, the navigation arrow on display no longer makes unrealistic jumps or suddenly changes orientation on its path.

To generate high-precision data of vehicle's motion, two tiny MEMS sensors—one for vehicle's acceleration and other for its yaw rate—have been housed into a single IC. The strength of the SMI230 lies in this combination. Also, the sensor duo achieves an extraordinary degree of precision and can detect changes in the vehicle's motion quickly.

Bosch

www.bosch.com

Sensor IC

Melexis, a provider of microelectronic solutions, has launched MLX90395 Triaxis Magnetometer Node, a monolithic sensor that uses the Hall effect for contactless 3D sensing. The MLX90395 meets the needs of demanding scenarios such as gear lever position sensing in automotive applications.

The functionality of the MLX90395 takes place through the system processor. It offers both I2C and SPI interfaces, making it simple to integrate into an automotive or industrial control environment.

The sensing solution's digital output provides a 16-bit resolution for X, Y, and Z magnetic field (Bx, By, and Bz) measurements, thus enabling host processor,



DSP, micro-controller, or digital signal controller to decode the

absolute position of any magnet as it passes the sensor.

The MLX90395 also has an integrated temperature sensor and supply voltage monitor. It operates in one of the three modes: single measurement mode, burst mode, or wake-up on change mode. The MLX90395 is smaller and more power-efficient than alternative Hall effect sensors.

Melexis

www.melexis.com

MISCELLANEOUS

Body cooling device

Due to rising outside temperatures and working in places with no room cooling facilities, maintaining our body temperature at an optimal level becomes a challenge.

To tackle this challenge, Sony has come up with an innovative pocket



air-conditioner

called Reon

Pocket. This

wearable air-

conditioner pro-

vides effective cool-

ing at any time of the day. In

fact, the device can also be used as a heater during winters.

The device works using the Peltier effect. It comes with a top-body innerwear that has a storage pocket at the back of the neck for holding the Reon Pocket. Whenever you need to use the device, simply wear the innerwear with the device firmly secured in the back pocket. Connect it to the Reon Pocket app on your smartphone via Bluetooth. It supports both Android and iOS functionalities.

Sony

www.sony.co.in